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Comparative Study of Metaheuristic Optimization Algorithms

Slide 1 — Title

Target Time: ~20 seconds

Good evening, everyone.

Our topic is a comparative study of **metaheuristic optimization algorithms**, specifically **Simulated Annealing, Tabu Search, and Particle Swarm Optimization**.

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We will compare them across three practical application areas: **route optimization, job scheduling, and machine learning feature selection**.

These were chosen because they are real-world problems where finding the perfect solution can be too slow or expensive, so good approximate solutions become valuable.

Slide 2 — Overview

Target Time: ~20 seconds

Here is what we will cover.

First, we will explain what optimization is and why exact search becomes difficult.

Second, we will introduce the three algorithms.

Third, we will discuss the three practical applications

Finally, we will present our research question, evaluation framework, and study plan.

Slide 3 — What is Optimization?

Target Time: ~35 seconds

Optimization means choosing the **best solution** from many possible options.

A solution is judged using an **objective function**, which measures how good or bad that solution is.

For example, in route optimization, the objective is usually to minimize distance. In job scheduling, it may be to minimize total completion time. In feature selection, it may be to improve model accuracy while using fewer features.

The challenge is that real-world problems can have too many possible solutions to check one by one. This is where metaheuristic algorithms become useful.

Slide 4 — Why Exact Search Becomes Difficult

Target Time: ~35 seconds

Some problems are simple to describe but difficult to solve exactly.

For example, in the **Traveling Salesman Problem**, the goal is to find the shortest route that visits every location once and returns to the starting point.

With only a few locations, checking all routes may still be possible. But as more locations are added, the number of possible routes grows extremely fast.

The same issue appears in job scheduling and feature selection. There can be too many possible schedules or feature subsets to test exhaustively.

Metaheuristics do not guarantee the perfect solution, but they aim to find a very good solution within a practical amount of time.

Slide 5 — What is a Metaheuristic?

Target Time: ~35 seconds

A **metaheuristic** is a smart search method for difficult optimization problems.

It searches large solution spaces without checking every possible answer.

Metaheuristics try to balance two goals: **exploration**, which means searching new areas, and **exploitation**, which means improving solutions that already look promising.

They are useful because they can avoid getting stuck in a local optimum, or a solution that seems good nearby but is not the best overall.

However, they still need to be adapted to each problem, which is why comparing them across applications is important.

Slide 6 — Three Algorithms at a Glance

Target Time: ~35 seconds

The three algorithms we are comparing are:

Simulated Annealing, which makes small changes to a solution and sometimes accepts worse results early on so it can escape local optima.

Tabu Search, which improves a solution while remembering recent moves so it does not repeat the same mistakes and cycling.

Particle Swarm Optimization, which uses a group of possible solutions that move based on their own best result and the best result found by the group.

All three solve optimization problems, but they search in very different ways.

Slide 7 — Simulated Annealing

Target Time: ~45 seconds

Simulated Annealing is inspired by heating and slowly cooling metal.

In optimization, it starts with one solution and makes small changes. If a new solution is better, it accepts it. If it is worse, it may still accept it early in the search which is what makes it different.

This helps the algorithm escape local optima instead of getting stuck too early.

Over time, the “temperature” decreases, so the algorithm becomes more selective and focuses on improving the best solution found.

For our study, Simulated Annealing can be used in all three applications: changing routes in the travelling salesman problem, adjusting schedules in job scheduling, and adding or removing features in machine learning.

Its main strength is that it is simple and flexible. Its main weakness is that its performance depends heavily on the cooling schedule.

Slide 8 — Tabu Search

Target Time: ~45 seconds

Tabu Search is also a single-solution search method, but its main feature is **memory**.

It keeps a tabu list of recent moves so the algorithm does not keep reversing itself or going in circles.

At each step, it checks nearby solutions and chooses the best move that is not tabu. If a forbidden move gives the best result so far, it can still be accepted through the aspiration criterion.

For our study, Tabu Search is especially useful for route optimization and job scheduling because those problems have clear move structures, such as swapping cities or reassigning tasks.

Its main strength is systematic memory. Its main weakness is that it can be more complex to tune and implement.

Slide 9 — Particle Swarm Optimization

Target Time: ~45 seconds

Particle Swarm Optimization is different because it uses a group of solutions instead of improving only one solution at a time.

Each solution is called a particle. Each particle remembers its own best result and also follows the best result found by the swarm.

This makes Particle Swarm Optimization good at exploring many areas at once.

For our study, Particle Swarm Optimization is most naturally suited to machine learning feature selection, especially using binary Particle Swarm Optimization, where each feature is either selected or not selected.

It can also be adapted for the travelling salesman problem and scheduling, but those require special encoding.

Its main strength is fast population-based search. Its main weakness is premature convergence, where particles cluster too early around a local optimum.

Slide 10 — Practical Application 1: TSP / Route Optimization

Target Time: ~40 seconds

The first practical application is **TSP route optimization**.

The goal is to find the shortest route that visits all locations once and returns to the starting point.

This applies to delivery routing, courier planning, and logistics.

For the experiment, all three algorithms can be tested on the same set of cities.

We can compare them based on route distance, runtime, convergence behavior, consistency, and ease of implementation.

SA and Tabu Search are natural fits because they can directly manipulate routes. PSO can also be tested, but it requires a discrete route encoding.

Slide 11 — Practical Application 2: Job Scheduling

Target Time: ~40 seconds

The second practical application is **job scheduling**.

The goal is to assign jobs to machines or workers so that all tasks are completed as efficiently as possible.

A common measure is **makespan**, which means the total time needed to finish all jobs.

This applies to factory scheduling, employee shifts, project planning, and machine task assignment.

For the experiment, each algorithm can be given the same jobs, processing times, and machines.

We can compare the final makespan, runtime, scalability, and tuning sensitivity.

Slide 12 — Practical Application 3: Machine Learning Feature Selection

Target Time: ~45 seconds

The third practical application is **machine learning feature selection**.

Feature selection means choosing which input variables should be used in a model.

The goal is to keep useful features while removing unnecessary or noisy ones. This can improve accuracy, reduce training time, and make the model easier to understand.

For the experiment, each solution can be represented as a binary string, where **1 means the feature is selected** and **0 means it is not selected**.

We can compare model accuracy, number of selected features, fitness score, runtime, and stability across runs.

This is where PSO may be especially useful, while SA and Tabu Search can also work by flipping features on or off.

Slide 13 — Research Question

Target Time: ~40 seconds

Given our coverage, our research question is:

How do Simulated Annealing, Tabu Search, and Particle Swarm Optimization differ in their effectiveness across route optimization, job scheduling, and machine learning feature selection in terms of solution quality, convergence speed, scalability, and ease of implementation?

Our supporting questions are:

1. How do the algorithms compare on TSP route optimization?
 2. How do they compare in job scheduling using makespan?
 3. How do they compare in feature selection using predictive performance and feature reduction?
 4. How does performance change as problem size increases?
 5. How difficult is each algorithm to adapt, tune, and implement?
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Slide 14 — Evaluation Framework

Target Time: ~45 seconds

Our evaluation framework uses the same criteria across the three applications, but each criterion is measured differently depending on the problem.

For **solution quality**, TSP uses route distance, job scheduling uses makespan, and feature selection uses model accuracy with fewer features.

For **convergence behavior**, we look at how quickly each algorithm improves.

For **scalability**, we look at what happens as the number of cities, jobs, machines, or features increases.

We also compare **parameter sensitivity**, or how much tuning affects the results.

Finally, we compare **implementation complexity**, meaning how difficult each algorithm is to adapt and run for each application.

Slide 15 — Study Scope and Plan

Target Time: ~35 seconds

The study will mainly compare existing research on Simulated Annealing, Tabu Search, and PSO.

We will look at how these algorithms are used in route optimization, job scheduling, and machine learning feature selection.

If feasible, we will also run small sample tests for each problem to support the comparison.

The final output will be a **Comparative Evaluation Framework** that helps answer:

Which algorithm should be used for which practical optimization problem?

The study does not assume one universal winner. Instead, it maps each algorithm's strengths to specific application contexts.

Slide 16 — Summary / Closing

Target Time: ~30 seconds

To summarize, our study compares Simulated Annealing, Tabu Search, and Particle Swarm Optimization across three practical optimization problems

These represent logistics, resource allocation, and artificial intelligence applications.

The goal is not to declare one algorithm universally best. The goal is to understand which algorithm is most suitable for each type of problem.

By comparing solution quality, convergence, scalability, tuning sensitivity, and implementation complexity, the study aims to produce a practical framework for algorithm selection.

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So thats all from us today, thank you for listening.